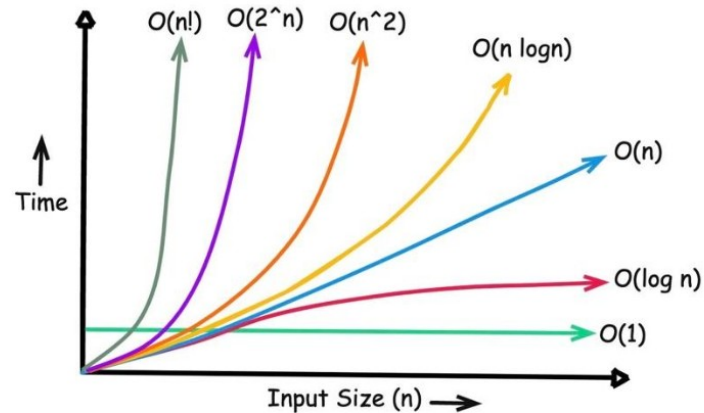


Lecture 7

Theoretical Running Time II



Announcements

- ❖ **Office hours** are up on course website
 - T: 12pm-4pm; F: 11am-1pm
 - Schedule an appointment individually or as a group
- ❖ **Lab 1** due this Friday @ 11:59pm
- ❖ **Project 1** Planning & Analysis due **next Friday** @ 11:59pm
 - Details up now on course website
 - Choose a problem and [join a team](#) if you haven't
 - Be sure to comment your Github link on the project document

Today

- ❖ Properties of Asymptotic Upper Bounds
- ❖ How to use Big-O in practice
- ❖ Examples

Big-O Definition

Let $T(n)$ be the number of steps required for our algorithm in the worst case

Definition: We say that $T(n)$ is $O(f(n))$ if there exists a $c > 0$ such that

$$T(n) \leq cf(n)$$

for sufficiently large n

Big-O Definition

Let $T(n)$ be the number of steps required for our algorithm in the worst case

Definition: We say that $T(n)$ is $O(f(n))$ if there exists a $c > 0$ such that

$$T(n) \leq cf(n)$$

for sufficiently large n

- ❖ Then f is an **asymptotic upper bound** of T

Common Bounds

- ❖ Constant time: $O(1)$
- ❖ Log time: $O(\log n)$
- ❖ Linear time: $O(n)$
- ❖ Log Linear time: $O(n \log n)$
- ❖ Quadratic time: $O(n^2)$
- ❖ Cubic time: $O(n^3)$
- ❖ Polynomial time: $O(n^k)$ for some $k > 0$
- ❖ Exponential time: $O(2^n)$

Faster Running Time
Slower Growth



Slower Running Time
Faster Growth

Examples

- ❖ If $T(n) = n^2 + 10000000n$, then $T(n)$ is $O(n^2)$

Examples

❖ If $T(n) = n^2 + 10000000n$, then $T(n)$ is $O(n^2)$

❖ If $T(n) = n \log n + n^3$, then $T(n)$ is $O(n^3)$

Examples

- ❖ If $T(n) = n^2 + 1000000n$, then $T(n)$ is $O(n^2)$
- ❖ If $T(n) = n \log n + n^3$, then $T(n)$ is $O(n^3)$
- ❖ If $T(n) = n + 6$, then $T(n)$ is **not** $O(1)$

Exercise 1

Let $T(n) = 3n + 17 \log n + 1000$. Which of the following are true?

Hint: More than one may be true.

a) $T(n)$ is $O(n^2)$

b) $T(n)$ is $O(n)$

c) $T(n)$ is $O(\log n)$

Exercise 1

Let $T(n) = 3n + 17 \log n + 1000$. Which of the following are true?

Hint: More than one may be true.

a) $T(n)$ is $O(n^2)$

b) $T(n)$ is $O(n)$

c) $T(n)$ is $O(\log n)$

Big-O bounds are upper bounds and do not need to be "tight"

Talking about Big-O

Suppose $T(n)$ is $O(n^2)$

Accurate things to say

- ❖ My algorithm has quadratic running time
- ❖ The worst case running time of my algorithm is $T(n)$

Sketchy thing to say

- ❖ $O(n^2)$ is **the** running time of my algorithm

How to use Big-O

- ❖ Study pseudocode to determine running time $T(n)$ for an algorithm as a function of n :

$$T(n) = 2n^2 + n + 2$$

- ❖ Prove that $T(n)$ is upper-bounded by a simpler function using the Big-O definition
 - Last time: showed that this $T(n)$ is $O(n^2)$
 - Simplify this reasoning by using properties of Big-O notation

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$

Examples

- $T(n) = n$ is linear time

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$
- ❖ Dropping multiplicative constants

Examples

- $T(n) = 5n$ is $O(n)$
- $T(n) = 3 \log n$ is $O(\log n)$
- $T(n) = 999999$ is $O(1)$
- $T(n) = 3 \cdot 2^n$ is $O(2^n)$

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$
- ❖ Dropping multiplicative constants
- ❖ Summing is a max

Example

- $T(n) = T_1(n) + T_2(n)$
- $T_1(n) = 5n$ is $O(n)$
- $T_2(n) = 3\log n$ is $O(\log n)$
- $T(n)$ is $O(\max\{n, \log n\})$
- $T(n)$ is $O(n)$

Allows us to only focus on
dominant terms w/r/t growth rate

Common Bounds

- ❖ Constant time: $O(1)$
- ❖ Log time: $O(\log n)$
- ❖ Linear time: $O(n)$
- ❖ Log Linear time: $O(n \log n)$
- ❖ Quadratic time: $O(n^2)$
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Faster Running Time
Slower Growth



Slower Running Time
Faster Growth

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$
- ❖ Dropping multiplicative constants
- ❖ Summing is a max
- ❖ Multiplication is a multiplication

Example

- $T(n) = T_1(n) * T_2(n)$
- $T_1(n) = 5n$ is $O(n)$
- $T_2(n) = 3 \log n$ is $O(\log n)$
- $T(n)$ is $O(n \log n)$

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$
- ❖ Dropping multiplicative constants
- ❖ Summing is a max
- ❖ Multiplication is a multiplication
- ❖ Polynomial degree grows faster

Example

- Suppose $0 < a < b$
- Then, n^a is $O(n^b)$

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$
- ❖ Dropping multiplicative constants
- ❖ Summing is a max
- ❖ Multiplication is a multiplication
- ❖ Polynomial degree grows faster
- ❖ Log is slower than polynomial

Example

- $\log n$ is $O(n^a)$ for any $a > 0$

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$
- ❖ Dropping multiplicative constants
- ❖ Summing is a max
- ❖ Multiplication is a multiplication
- ❖ Polynomial degree grows faster
- ❖ Log is slower than polynomial
- ❖ Exp is faster than polynomial

Example

- n^a is $O(2^n)$ for any $a > 0$

Properties

- ❖ Reflexivity: $T(n)$ is $O(T(n))$
- ❖ Dropping multiplicative constants
- ❖ Summing is a max
- ❖ Multiplication is a multiplication
- ❖ Polynomial degree grows faster
- ❖ Log is slower than polynomial
- ❖ Exp is faster than polynomial
- ❖ Transitivity

Example

- Suppose $f(n)$ is $O(g(n))$ and $g(n)$ is $O(h(n))$, then $f(n)$ is $O(h(n))$
- $5n$ is $O(n)$ and n is $O(n^2)$ then $5n$ is $O(n^2)$

Examples

- ❖ If $T(n) = n^2 + 1000000n$, then $T(n)$ is $O(n^2)$
- ❖ If $T(n) = n \log n + n^3$, then $T(n)$ is $O(n^3)$
- ❖ If $T(n) = n + 6$, then $T(n)$ is **not** $O(1)$

Next Time

- ❖ Asymptotic Lower & Exact Bounds
- ❖ DSP 5.4-5.10